

Theory: Two-Spin Quantum Simulation Using Qiskit

Introduction

One of the most important applications of quantum computing is the simulation of quantum systems. Classical computers often struggle to simulate large quantum systems because the amount of information grows exponentially with the number of particles. Quantum computers, however, naturally operate according to the laws of quantum mechanics and can therefore represent quantum systems more efficiently.

This project demonstrates a simple example of quantum simulation using two qubits. The two qubits represent two interacting quantum spins. By preparing an entangled quantum state and measuring its properties, we can study the behavior of a small quantum system using Qiskit.

Quantum Spins

A spin is a fundamental quantum property of particles such as electrons. For simplicity, a spin can be viewed as a tiny magnet that can point in one of two directions:

Spin Up $\rightarrow |0\rangle$
Spin Down $\rightarrow |1\rangle$

Unlike classical objects, quantum spins can also exist in a superposition of both states simultaneously.

For a single qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

$$|\alpha|^2 + |\beta|^2 = 1$$

The coefficients determine the probabilities of measuring the qubit in each state.

Quantum Entanglement

When two qubits become correlated in a way that cannot be explained classically, they are said to be entangled.

In this project, we prepare the Bell state:

$$|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$$

This state has an important property:

- If the first qubit is measured as 0, the second will be 1.
- If the first qubit is measured as 1, the second will be 0.

The measurement outcomes are perfectly correlated even though neither qubit has a definite value before measurement.

Creating the Bell State

The circuit begins with two qubits initialized in:

$$|00\rangle$$

Step 1: Hadamard Gate

Applying a Hadamard gate to the first qubit creates a superposition:

$$|00\rangle \rightarrow (|00\rangle + |10\rangle)/\sqrt{2}$$

Step 2: CNOT Gate

The CNOT gate entangles the two qubits:

$$\begin{aligned} &(|00\rangle + |10\rangle)/\sqrt{2} \\ &\rightarrow \\ &(|00\rangle + |11\rangle)/\sqrt{2} \end{aligned}$$

This state is the Bell state:

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$$

Step 3: Additional Gates

Applying an X gate to the second qubit and a Z gate to the first qubit transforms the state into:

$$|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$$

which is the state used throughout this project.

Measuring the Quantum State

After preparing the Bell state, measurements are performed on both qubits.

The theoretically expected outcomes are:

01 $\approx$ 50%

10 $\approx$ 50%

The outcomes:

00

11

should ideally never appear.

Because real quantum computers are affected by noise and hardware imperfections, small deviations from the theoretical distribution may occur.

The Two-Spin Hamiltonian

To study the energy of the system, we define a Hamiltonian.

A Hamiltonian is a mathematical operator that describes the total energy of a quantum system.

For this project:

$$H = J(ZZ) + \hbar\omega(XI + IX)$$

where:

J = interaction strength between spins

$\hbar\omega$ = external transverse magnetic field

Meaning of Each Term

Interaction Term

$$J(ZZ)$$

describes how the two spins interact with each other.

Depending on the value of J , the spins may prefer to align or anti-align.

Magnetic Field Term

$$\hbar\omega(XI + IX)$$

represents the effect of an external magnetic field acting on both spins.

This field tends to flip the spins and create quantum superpositions.

Measuring Energy with Estimator

In Qiskit, two different primitives are used in this project.

Sampler

The Sampler primitive answers the question:

What measurement outcomes occur and how often?

Output example:

01 : 50

10 : 49

11 : 1

The result is a probability distribution of measurement outcomes.

Estimator

The Estimator primitive answers a different question:

What is the expectation value of an observable?

Instead of producing measurement counts, it calculates a physical quantity such as energy.

For this project, Estimator evaluates:

$$\langle \Psi | H | \Psi \rangle$$

which represents the energy of the Bell state under the chosen Hamiltonian.

Circuit Optimization (Transpilation)

Before execution, the quantum circuit must be translated into instructions supported by the target quantum processor.

This process is called transpilation.

During transpilation:

- Logical qubits are mapped to physical qubits.
- Unsupported gates are replaced by hardware-native gates.
- Circuit depth may be reduced.
- Execution efficiency is improved.

Qiskit performs these tasks automatically using a preset pass manager.

Real Hardware vs Simulator

This project runs on both:

IBM Quantum Hardware

Advantages:

- Uses a real quantum processor.
- Demonstrates practical quantum computation.

Disadvantages:

- Queue waiting times.
 - Noise and hardware errors.
-

Aer Simulator

Advantages:

- Fast execution.
- No waiting time.
- Ideal theoretical results.

Disadvantages:

- Does not represent real hardware imperfections unless a noise model is added.
-

Results

The Bell state measurements should show that the two qubits are anti-correlated:

01 $\approx$ 50%

10 $\approx$ 50%

The energy calculated using Estimator should be close to:

-1

for the chosen Hamiltonian parameters.

Small differences may occur on real hardware due to noise and gate imperfections.

Conclusion

This project demonstrates several fundamental concepts in quantum computing:

- Qubits
- Superposition
- Quantum Gates
- Entanglement
- Bell States
- Quantum Measurement
- Hamiltonians
- Quantum Simulation
- Sampler Primitive
- Estimator Primitive
- IBM Quantum Hardware
- Aer Simulation

Although the system contains only two qubits, the same principles can be extended to larger quantum systems. Quantum simulation is expected to be one of the most impactful applications of quantum computers because it enables the study of physical systems that are difficult or impossible to model efficiently on classical computers.

Overview

This project demonstrates the fundamentals of quantum computing through the simulation of a simple two-spin quantum system using Qiskit.

The project explores:

- Quantum Superposition
- Quantum Entanglement
- Bell State Preparation
- Quantum Measurement using SamplerV2
- Energy Estimation using EstimatorV2
- IBM Quantum Hardware Execution
- Aer Simulator Execution
- Hamiltonian-Based Quantum Simulation

The goal is to understand how quantum computers can be used to simulate physical quantum systems and calculate observable quantities such as energy.

Project Motivation

One of the most promising applications of quantum computing is quantum simulation.

Classical computers face exponential scaling challenges when simulating quantum systems. Quantum computers naturally follow the laws of quantum mechanics and can therefore model these systems more efficiently.

This project simulates a simple two-spin system and demonstrates how quantum circuits can be used to study quantum states and measure physical properties.

Quantum State Used

The project prepares the Bell State:

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

This state is maximally entangled and exhibits perfect anti-correlation between the two qubits.

Hamiltonian

The energy of the system is described by:

$$H = J(ZZ) + h_x(XI + IX)$$

where:

Parameter Description

J	Spin-spin interaction strength
hx	Transverse magnetic field strength

This Hamiltonian models:

- Interaction between two spins
- Influence of an external magnetic field
- Quantum spin flipping effects

Qiskit Workflow

This project follows the standard Qiskit Patterns workflow:

1. Map

Represent the physical system using qubits and quantum operators.

2. Optimize

Transpile the circuit for the selected quantum backend.

3. Execute

Run the circuit using:

- IBM Quantum Hardware
- Aer Simulator

4. Post-Process

Analyze:

- Measurement probabilities
 - Energy expectation values
-

Experiments

Experiment 1: Bell State Measurement on IBM Quantum Hardware

Using SamplerV2:

Bell State → Transpile → IBM Quantum Backend → Counts

Expected Results:

01	≈	50%
10	≈	50%

Experiment 2: Bell State Measurement on Aer Simulator

Using BackendSamplerV2:

Bell State → Aer Simulator → Counts

Purpose:

- Fast execution
 - Hardware comparison
 - Noise-free validation
-

Experiment 3: Energy Estimation on IBM Quantum Hardware

Using EstimatorV2:

Bell State + Hamiltonian → Estimator → Energy

Energy calculated:

$$\langle \Psi | H | \Psi \rangle = \langle \Psi | H | \Psi \rangle$$

Experiment 4: Energy Estimation on Aer Simulator

Using BackendEstimatorV2:

Bell State + Hamiltonian → Simulator → Energy

Used to compare theoretical and experimental results.

Technologies Used

- Python 3.10+
- Qiskit
- Qiskit Aer
- Qiskit IBM Runtime
- IBM Quantum Platform
- Jupyter Notebook